

IDMR 2026 Hw2

CONVERSATIONAL AGENT

Assignment Objective

Overall

1. Design a scenario (e.g. study companion, doctor) that can integrates context-aware conversational AI.
2. Record a Demo to show how it works.
3. Write a report for your homework to describe in detail.

Spec of the Assignment

Things you must do :

Your editor version must be 6000.0.50f1 !

- Using higher or lower version may cause unknown bug

Create apps using your meta account and activate quest link

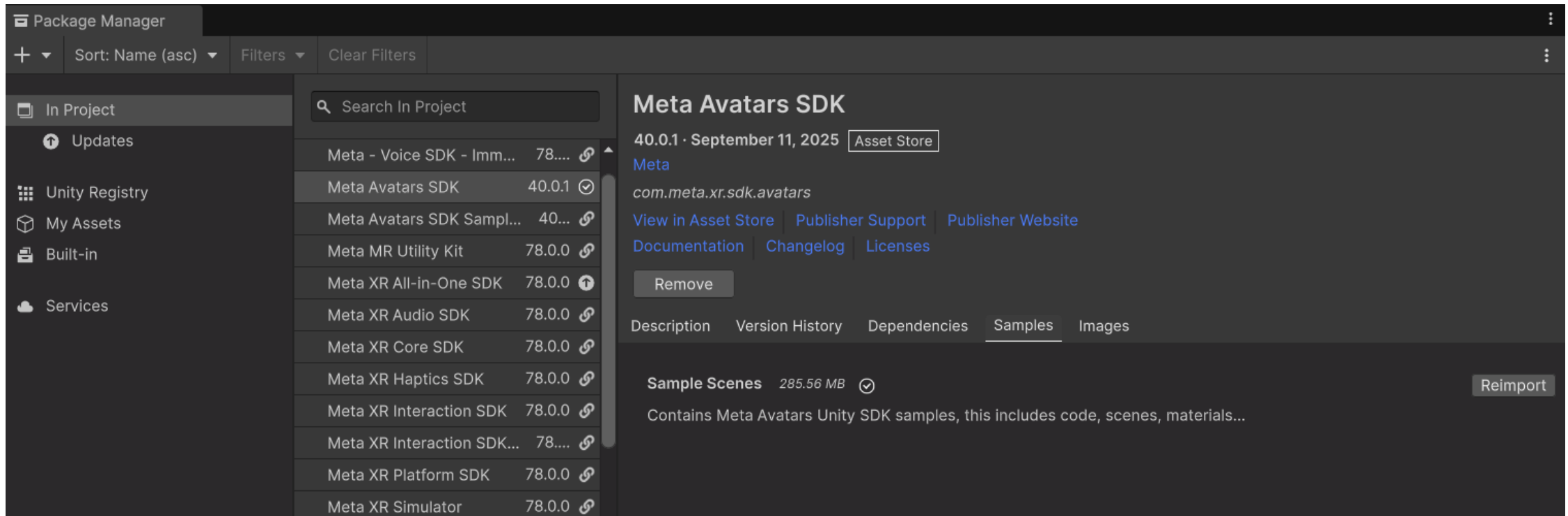
- You must register as a developer
- Open unknown source, developer setting in Quest link

Using the 'key' to access your LLM API

Reimport meta avatar sample

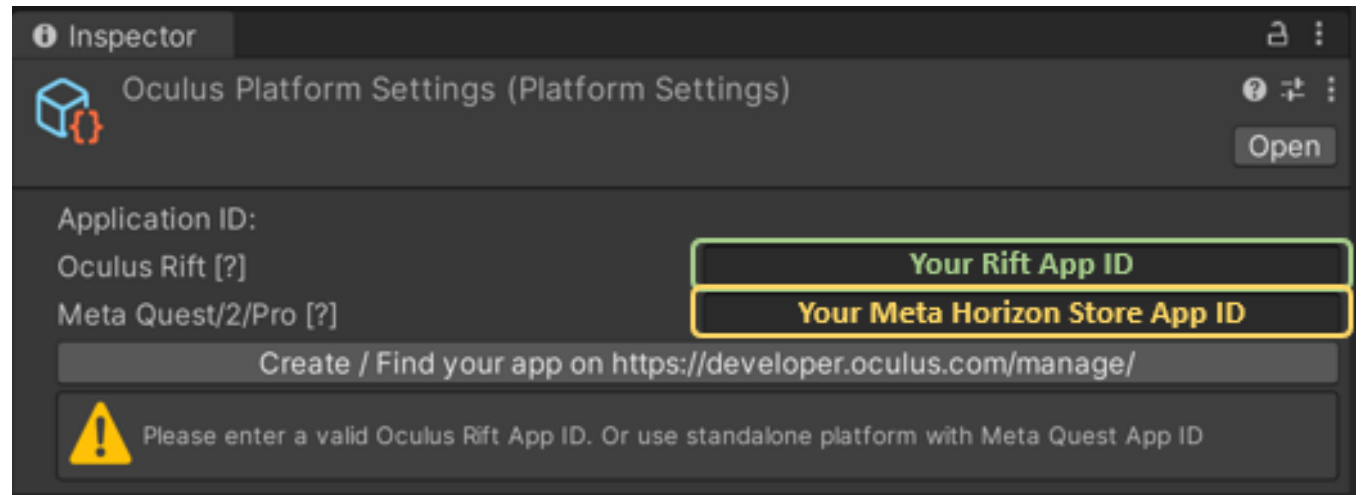
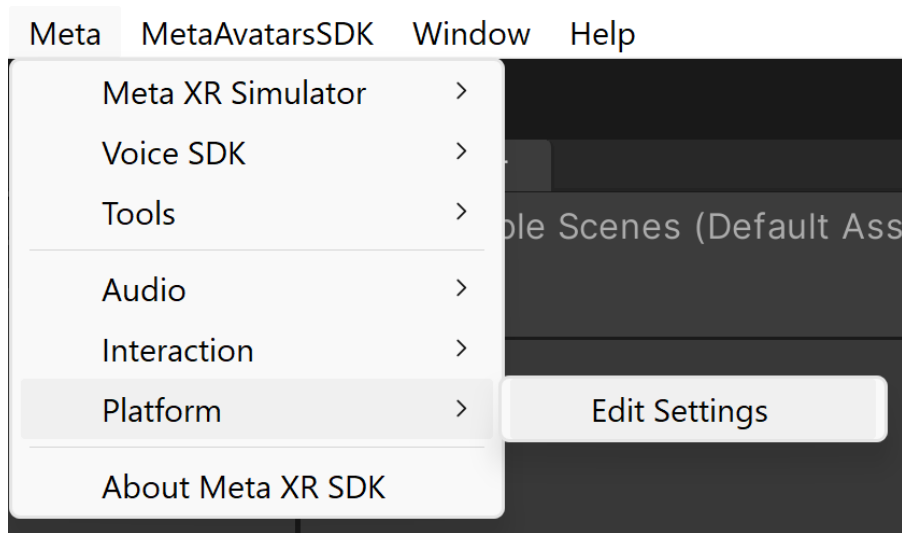
Spec of the Assignment

Remember to reimport sample of Avatar SDK



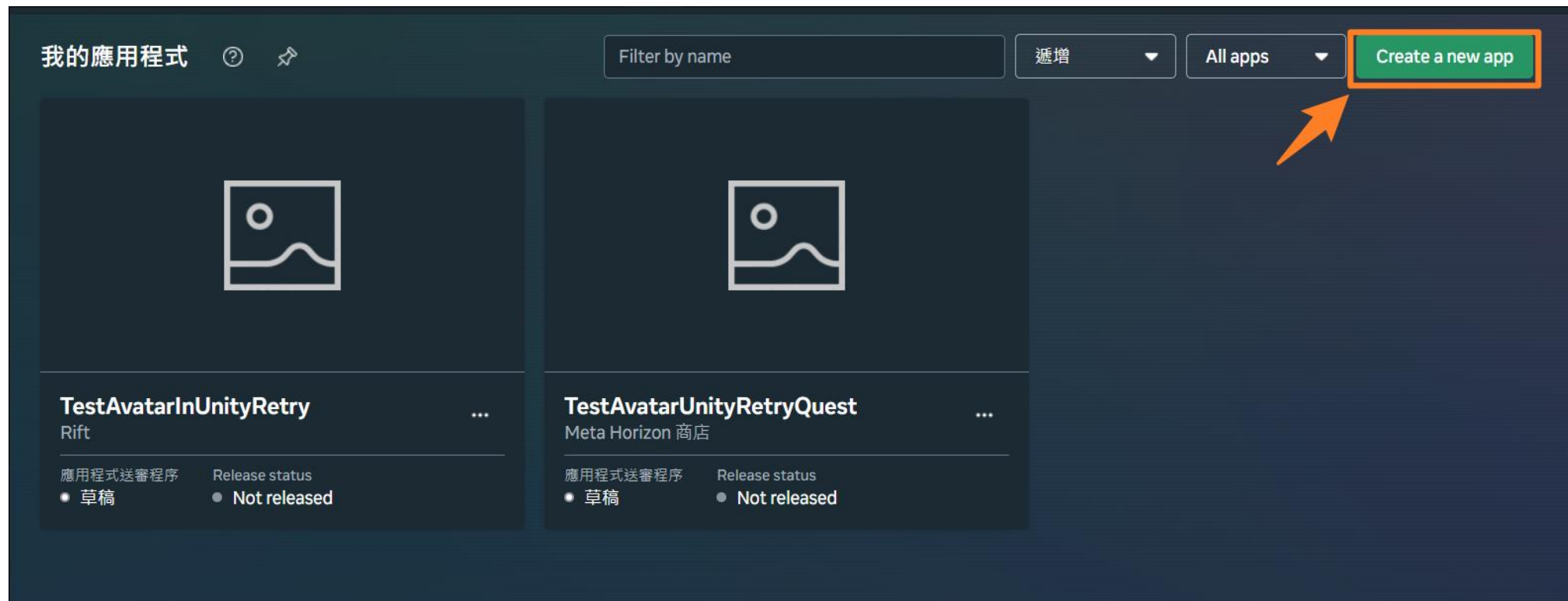
Spec of the Assignment

Paste your App ID to platform Inspector



Spec of the Assignment

Create app : Rift, Quest



Spec of the Assignment

Register as a developer

The screenshot displays the Meta Horizon Dashboard for a developer. The left sidebar contains navigation links: 'No app selected', 'App Overview', 'Development', 'Distribution', 'Growth', 'Engagement', 'Monetization', 'Analytics', 'Requirements', 'Settings', 'Organization', 'Account', 'Verification', 'Account Settings', and 'Organization Invites'. The main content area is titled '驗證你的 Meta Horizon 開發人員帳號' (Verify your Meta Horizon developer account). It includes a search bar, shortcuts, and a notification that the account is verified. Below this, there are sections for '設定雙重驗證' (Set up two-step verification), 'Add Mobile Number', and 'Add credit card', each with a '設定' (Set up) button and a 'Completed' status indicator.

Meta Horizon Dashboard

Search for pages or documentation

Shortcuts

No app selected
Select an app

App Overview

Development
Build your app

Distribution
Get feedback & release

Growth
Acquire more users

Engagement
Retain existing users

Monetization
Increase your revenue

Analytics
View your data

Requirements
Stay in compliance

Settings
Manage your app

Organization
PoYanWu

Account
brian910220

通知

Verification

Account Settings

Organization Invites

驗證你的 Meta Horizon 開發人員帳號

若要成為 Meta Horizon 開發人員，你必須設定雙重驗證或提供信用卡，以協助我們證實你的身分。如果移除這項資訊，你的帳號就不再是已驗證。瞭解詳情

你已成功驗證 Meta Horizon 開發人員帳號

返回首頁

設定雙重驗證
首先請點擊「設定」，讓系統將你導向 Meta 帳號管理中心，選擇雙重驗證並選擇你的 Meta 帳號，然後完成雙重驗證機制的設定。

設定

Add Mobile Number
新增手機號碼到 Facebook 帳號。
編輯手機號碼

Completed

Add credit card
This will be saved in your Meta account under payment methods. You can also use it for Meta Horizon Store purchases.
Edit payment information

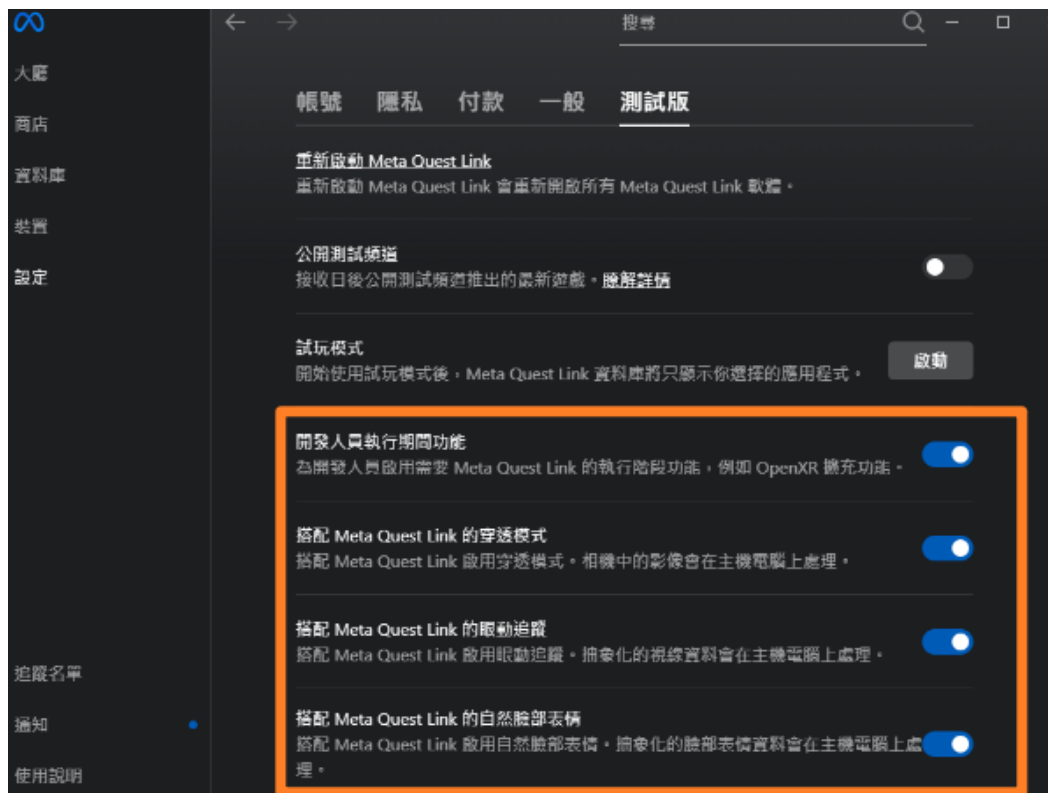
Completed

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隱私政策 法律聲明

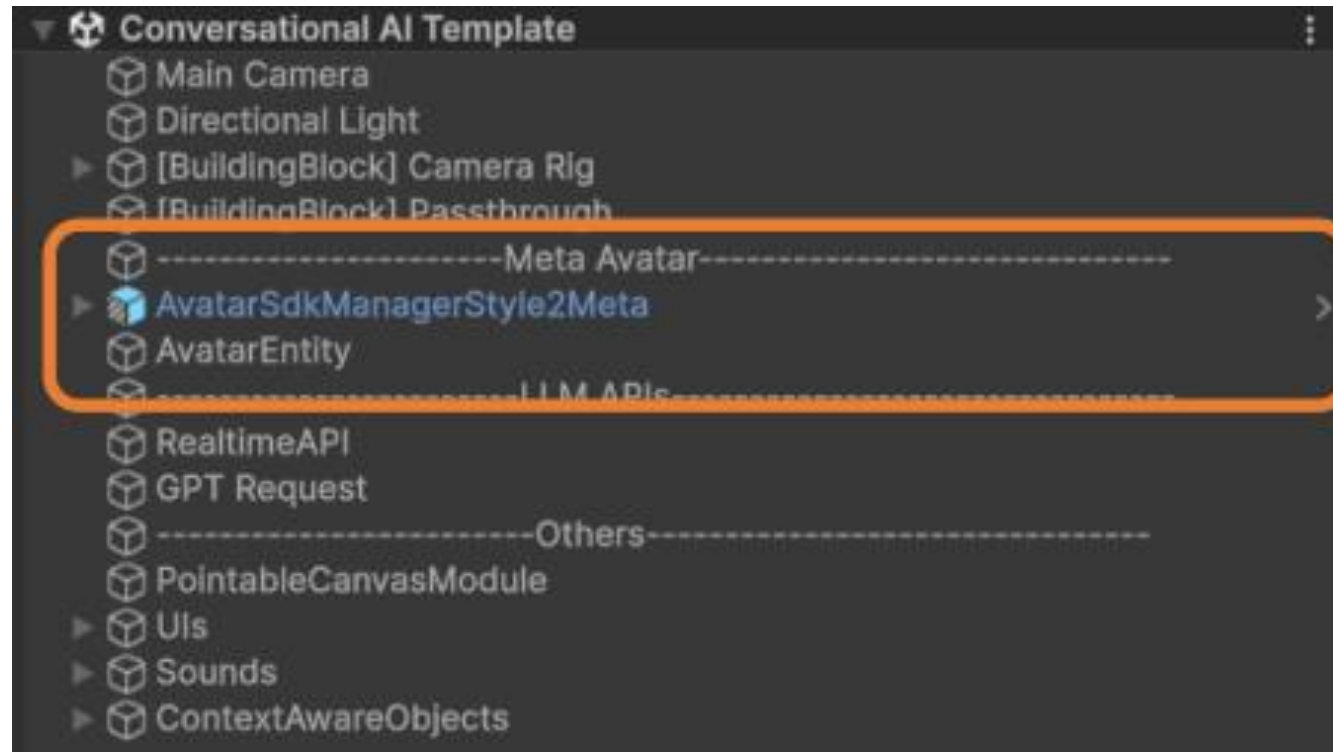
Spec of the Assignment

Remember to do if you switch your quest account to yours



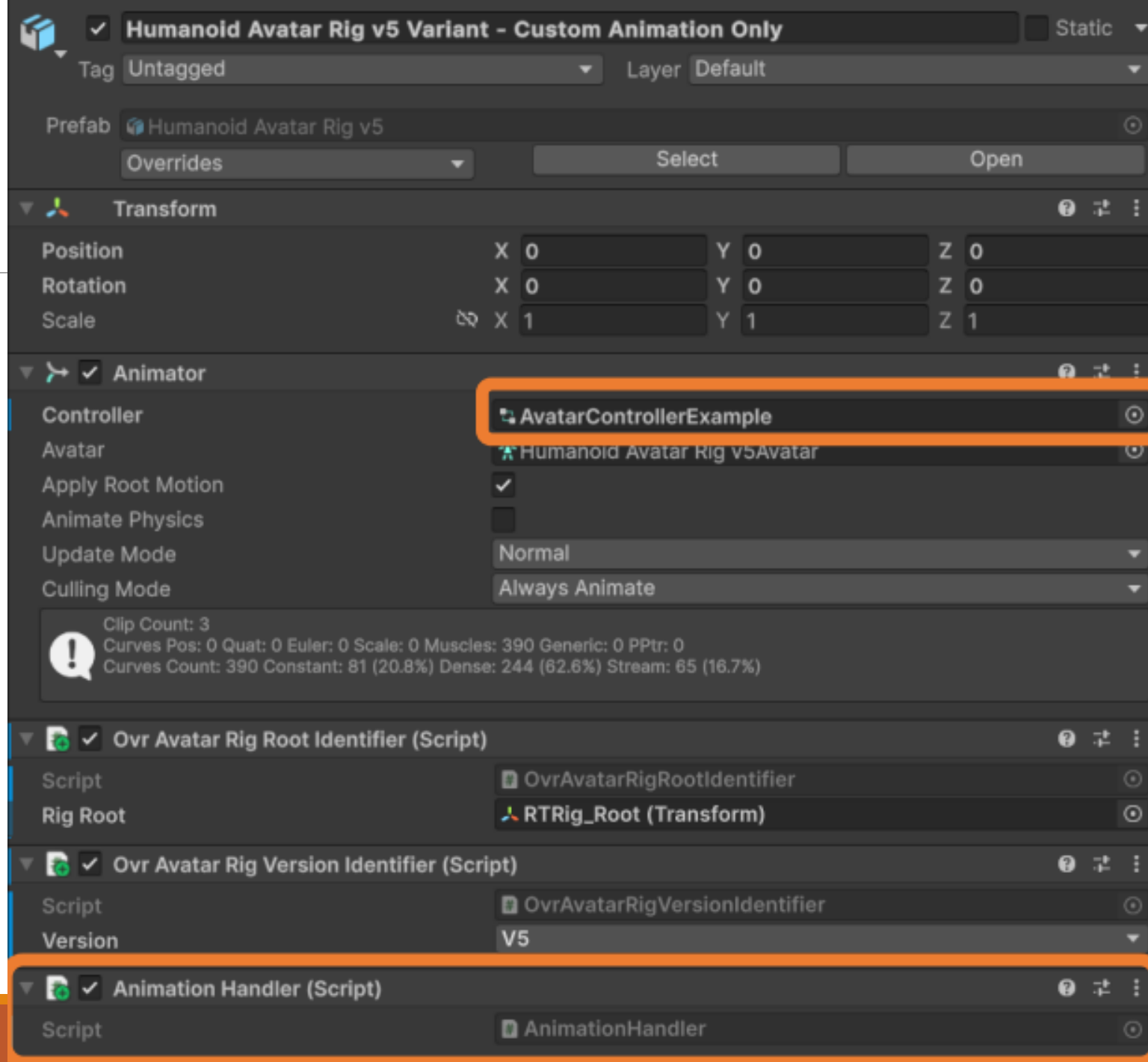
Scene obj of Assignment

Meta Avatar



Animation handler :

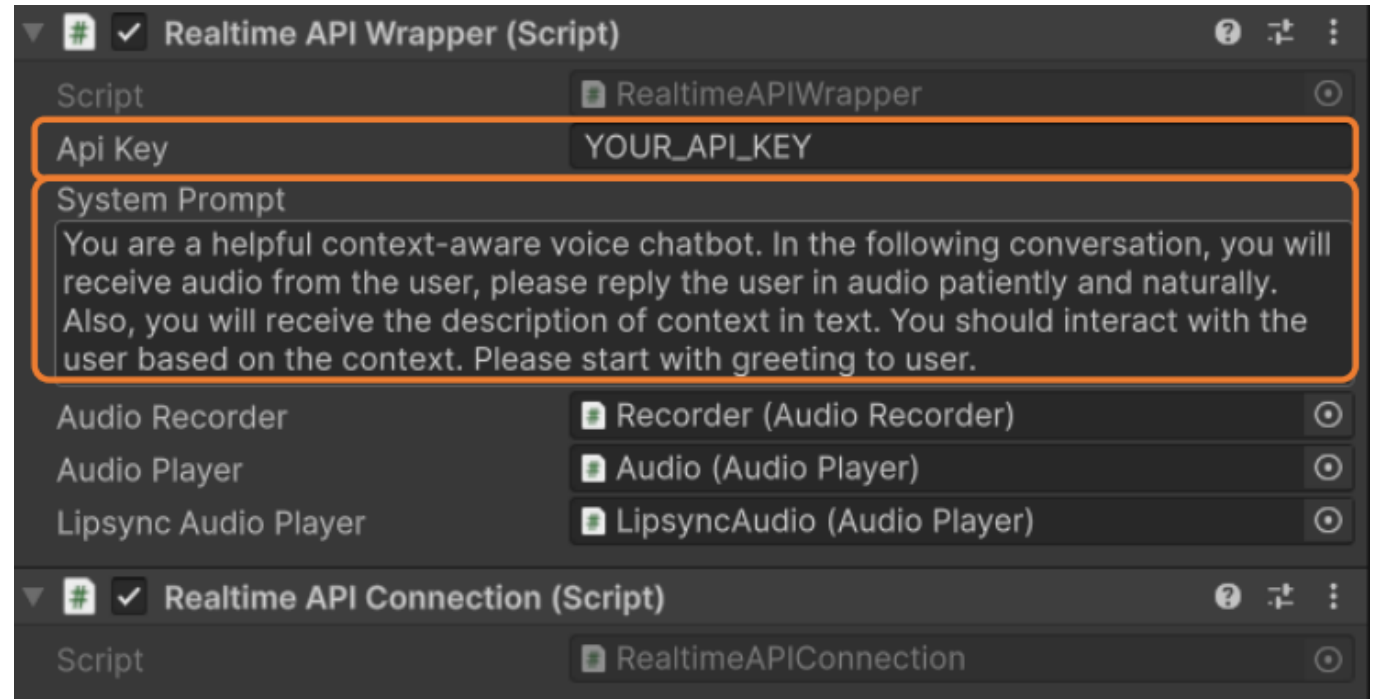
Control the animation by modifying the parameters in Animation controller



Scene obj of Assignment

Realtime API Wrapper

- Use websocket for Realtime API connection(4o-mini)
- API key
- System prompt

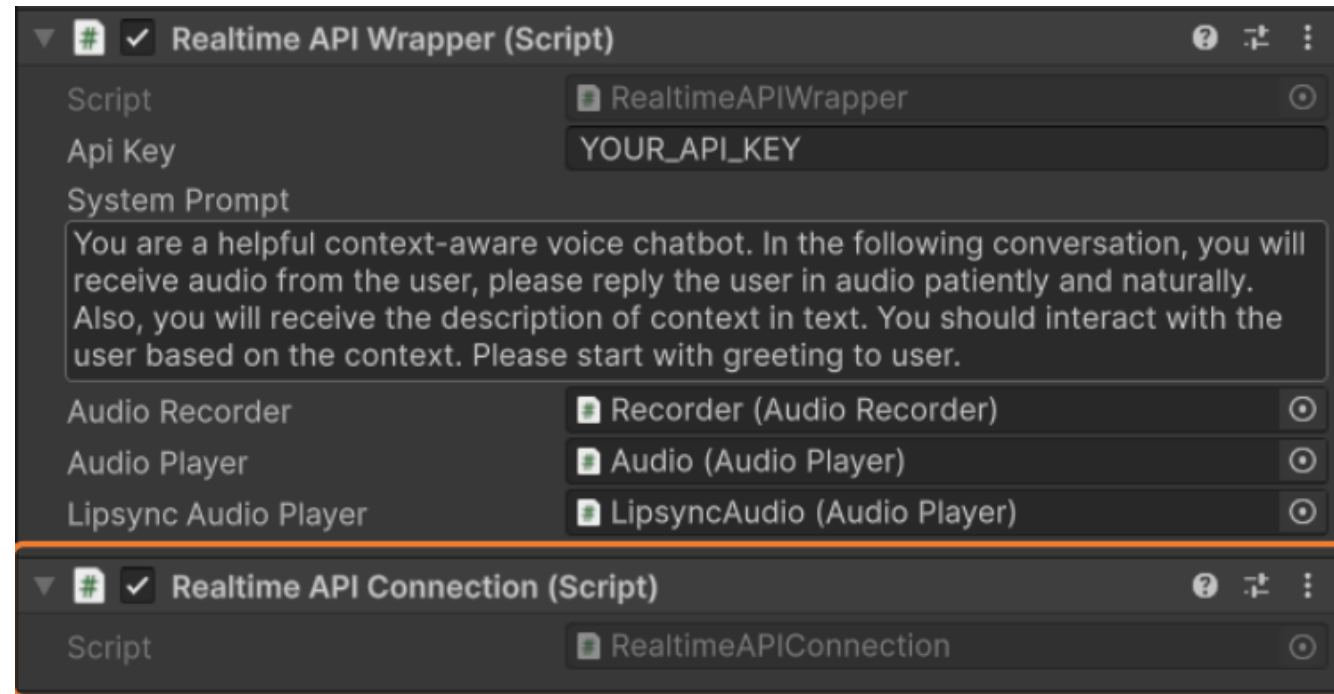
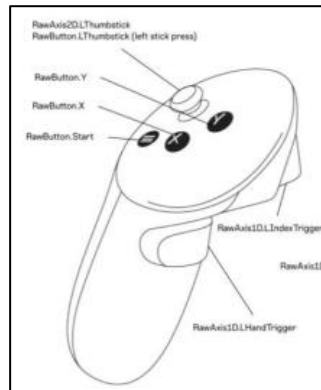
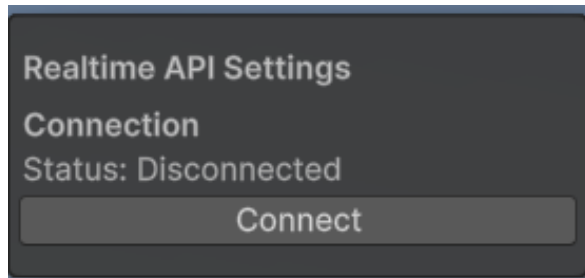


Scene obj of Assignment

Realtime API Connection : Connect to the socket only when needed
token are expensive!

Connect through :

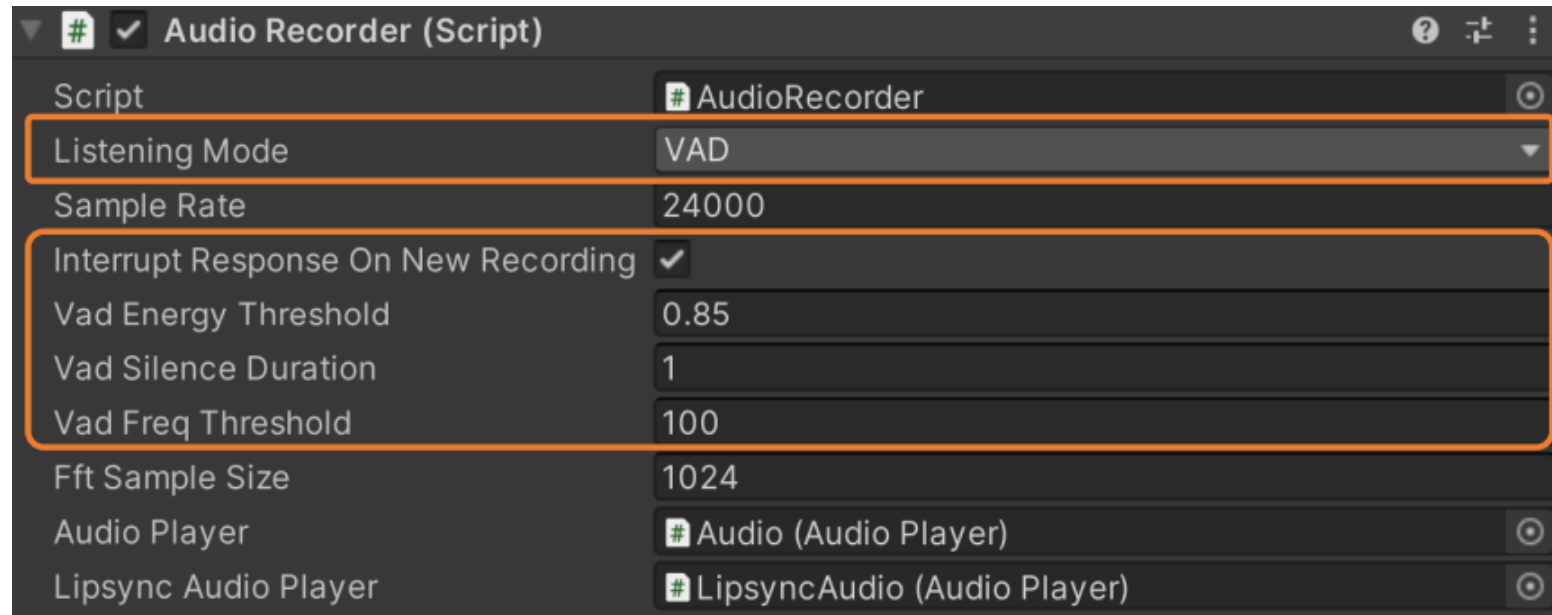
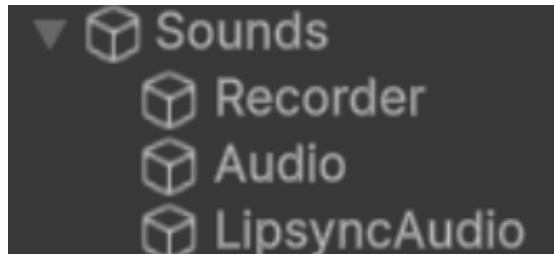
1. Scene GUI
2. Control Button Y



Scene obj of Assignment

Recorder :

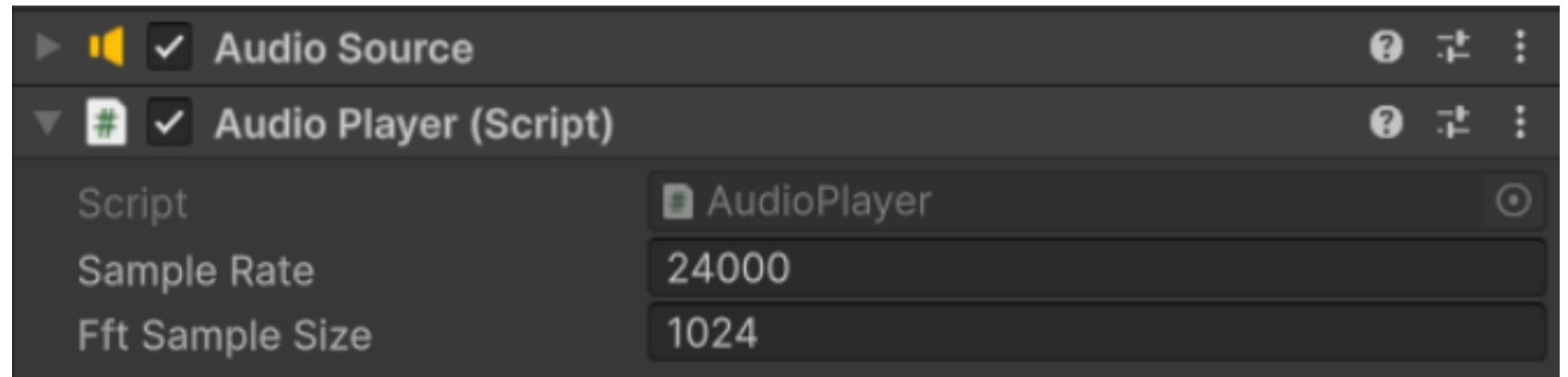
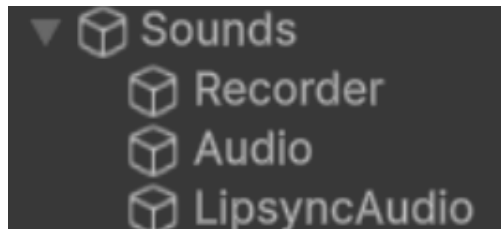
- Voice Activity Detection (VAD)



Scene obj of Assignment

Audio :

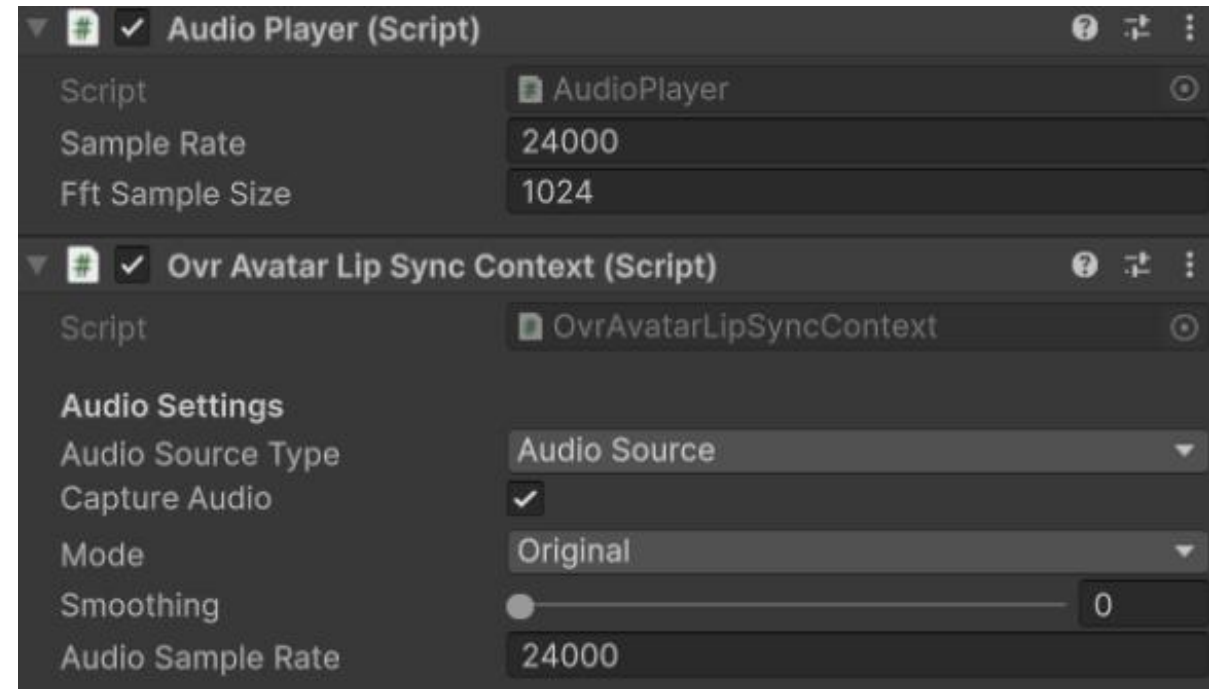
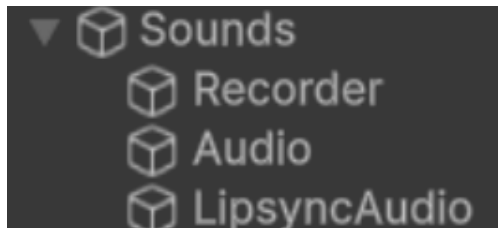
- Play the audio response from the Realtime API



Scene obj of Assignment

LipSync Audio :

- Stream the audio for avatar lip-sync



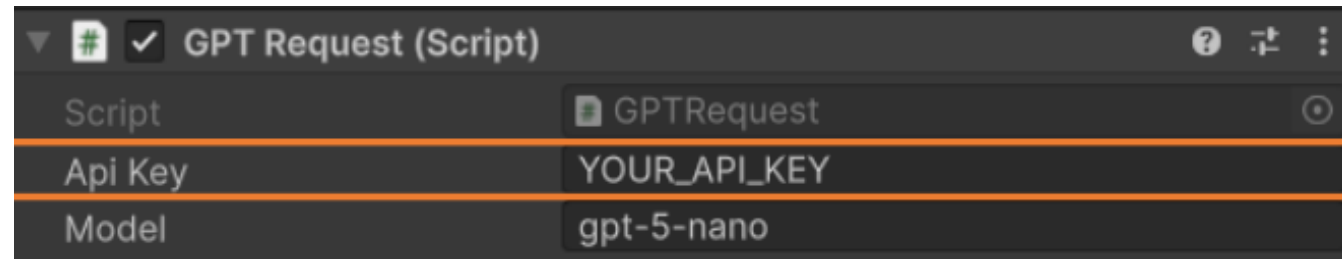
Scene obj of Assignment

GPT Requester (example) :

1. GPT Request
2. functionANDTool
3. structureRequest

Parameter :

1. API key
2. Model



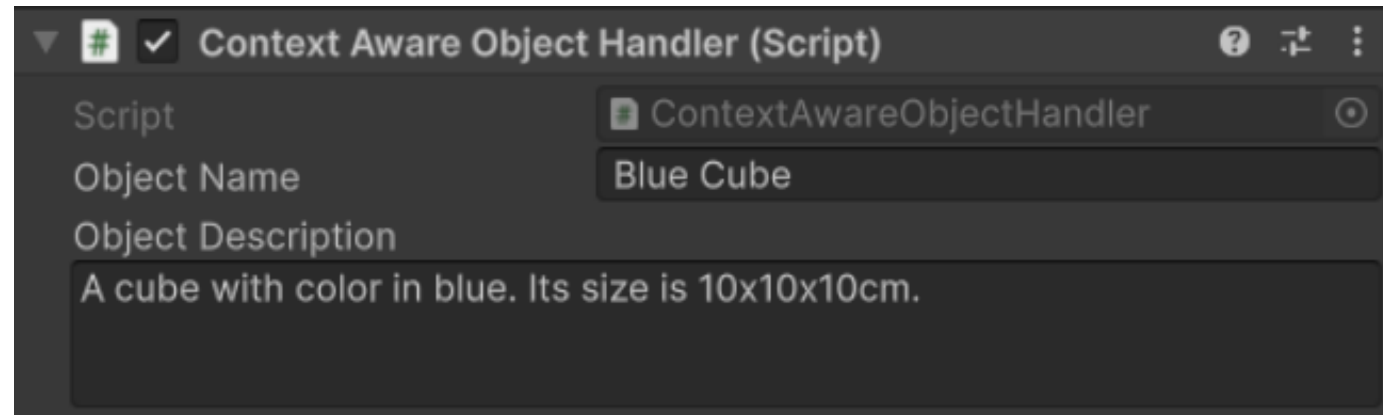
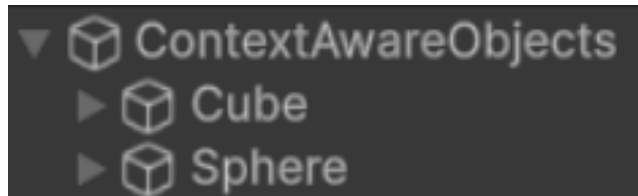
Scene obj of Assignment

Context aware object :

Just a Simple showcase, feel free to modify or remove it. We encourage you to use your creativity in the implementation.

Contains :

1. Name
2. Description



Spec of the Assignment

Scripts recommend you to watch (may insight your coding)

- RealtimeAPIWraper
- functionANDTool (example)
- GPTRrequest (example)

Assignment Objective

Project Goal - Basic

Replace the sample avatar with your own

- e.g. Custom meta avatar, or Ready Player Me
- If you choose Meta Avatar, you cannot just use sample skin provide by meta.

Add animatoins for your avatar.

- Include lip-sync
- 3 custom animation, triggered by conversation dialogues

Contex-Aware Interactoin

- Combine at least one virtual or physical context with the conversation.

Assignment Objective

Project Goal – Advance

More animations

Include more avatar dialogue for richer interactions (e.g. personalities, can trigger something according to the conversations...)

Richer UI

Interact with avatar through scene objects.

Anything special you can add !

Report structure

Things you must have in your report:

Your name, student id, demo video link, cloud file link of your project

For the rest, you can write your report follow the structure below (but not mandatory)

Motivation of your scenario

Implementation of basic goal

Advance goal (bonus) of your project.

Reflection

Assignment Grading

Basic + Demo Video (60%)

Report (10%)

Popularity (10%)

- Rated by peer evaluation

Advanced (20%)

- Technical, Aesthetics, Creative, Natural...etc

Late submission : -5% per day

Due date 5/23 23:59

FOR 2 WEEK + 1 DAY